

BOBICHENG ZHANG

PHD CANDIDATE · PSYCHOLOGY · COGNITIVE NEUROSCIENCE

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EDUCATION

Michigan State University

PHD PSYCHOLOGY

East Lansing, MI

Dec 2022 - Present

- Dissertation (Defended Dec 2025): Dissecting pupillary response pattern accompanying perceptual switches during bistability

Michigan State University

MA PSYCHOLOGY

East Lansing, MI

Jul 2020 - Dec 2022

- Thesis: Modest Effect of Knowledge on Bistable Perception during Structure-from-Motion

University of Chicago

MA SOCIAL SCIENCES

Chicago, IL

Oct 2018 - Jun 2019

- Thesis: Ambiguity resolution with multi-frequency interocular switch rivalry

University of Denver

BS PSYCHOLOGY

Denver, CO

Sep 2014 - Jun 2018

- Minors in Mathematics and Sociology

RESEARCH

Michigan State University - Department of Psychology

ADVISOR: JAN BRASCAMP

East Lansing, MI

Aug 2020-Present

- Collect and analyze physiological data such as eye movement, pupil size, skin conductance and heart rate.
- Develop custom algorithms in Python that allow for inference of perceptual interpretations based on eye movements during bistability without explicit report.
- Present findings at various conferences such as Society for Neuroscience and Vision Sciences Society.
- Advise and support undergraduate students in literature review, statistical analysis and presenting at conference.

Energize Colorado - Research and Insights

SUPERVISOR: DUSTIN WEILBACH

Denver, CO

Nov 2021 - Oct 2025

- Developed dozens of surveys using Qualtrics to collect information from small business owners to compute performance and health metrics.
- Performed extensive data wrangling for complex data structures to extract clearer insights and prepare for further analysis.
- Conducted exploratory analyses such as principle component analysis and parallel analysis to establish construct validity for hypothesized components of small business resilience.
- Collaborated with other teams at EC to maximize the utilization and effectiveness of actionable, data-driven insights.

University of Chicago - Institute for Mind and Biology

ADVISOR: STEVE SHEVELL

Chicago, IL

Oct 2018 - Jul 2020

- Reviewed literature, conducted experiments and completed a highly rated MA thesis in less than 8 months.
- Performed statistical analyses such as non-orthogonal contrast to demonstrate that binocularly driven neurons in V1 resolve neural ambiguity during chromatic interocular-switch rivalry.
- Presented findings at a graduate student conference and departmental forums.

University of Denver - Department of Psychology

ADVISOR: TIMOTHY SWEENEY

Denver, CO

Nov 2016 - Jun 2018

- Independent Study: "The effect of chromatic adaptation on affective perceptual decision making"
- Reviewed literature in efforts to understand relationships between colors and emotion perception.
- Learned to program psychophysics experiments in Matlab to present morphed emotional faces that were backward-masked.

PUBLICATIONS

Brascamp JW, **Zhang B**, Marshev V (2025) Fixational saccade inhibition and pupil dilation during self-paced limb movement preparation. *PLoS One* 20(10): e0335504. <https://doi.org/10.1371/journal.pone.0335504>

Zhang, B., Marshev, V., & Brascamp, J. W. (2025). The pupil response to perceptual switches: What happens when you ignore them. *Journal of Vision*, 25(8), 5-5. <https://doi.org/10.1167/jov.25.8.5>

Smith, O.M., Jeffery, B.E., Leuenberger, W., Mack, K.M, Pizzo, G, **Zhang, B** & Davis, C. Widespread lack of article accessibility policies among ecology and evolution journals. *Nat Ecol Evol* (2025). <https://doi.org/10.1038/s41559-025-02694-8>

Zhang, B. & Brascamp, J.W. (2022). Modest effect of knowledge on bistable perception of structure-from-motion. *Vision Research*. <https://doi.org/10.1016/j.visres.2022.108118>

Zhang, B., Slezak, E., Wang, W. & Shevell, S.K.(2021). Binocularly-driven competing neural responses and the perceptual resolution of color. *Journal of Vision*, 21(10), 15-15. <https://doi.org/10.1167/jov.21.10.15>

PRESENTATIONS

* *presenting author*; + *presented by lead author*

+Brascamp, J. W., **Zhang, B.**, & Marshev, V. (July 2025). Rate of fixational saccades reveals intention to move limbs. Poster Presentation, Gordon Research Conference, South Hadley, MA

+Cottone, M., Padula, I., Gampala, G. & Brascamp, J. W. (April 2025). Investigating the Impact of Stimulus Movement on Pupil Size Independent of Luminance. Poster Presentation, The Undergraduate Research and Arts Forum, East Lansing, MI

+Brascamp, J. W., **Zhang, B.**, Frey, H & Marshev, V. (November 2024). Habituation of the pupil response to visual events. Poster Presentation, The Psychonomic Society Annual Meeting, New York City, NY

***Zhang, B.**, Brascamp, J. W., (May 2024). Perceptual switches during structure-from-motion do not elicit pupil dilation. Poster Presentation, Vision Sciences Society, St. Pete's Beach, FL

+DuRussel, B., Karkare, T., Gampala, G. & Brascamp, J. W. (April 2024). Differences in effort-Related Pupil Responses between auditory and visual tasks. Poster Presentation, The Undergraduate Research and Arts Forum, East

Lansing, MI

- ***Zhang, B.**, Brascamp, J. W., (November 2023). Inferring perceptual switches during structure-from-motion without report. Poster Presentation, The Psychonomic Society Annual Meeting, San Francisco, CA
- ***Zhang, B.**, Brascamp, J. W., (August 2023). Separating event-related pupillary responses during perceptual bistability. Poster Presentation, Systems Vision Science Symposium, Tübingen, Germany
- ***Zhang, B.**, Brascamp, J. W., (November 2022). Modest effect of knowledge on bistable perception of structure-from-motion. Poster Presentation, Society for Neuroscience, San Diego, CA
- ***Zhang, B.**, Brascamp, J. W., (February 2022). Modest effect of knowledge on bistable perception of structure-from-motion. Oral Presentation, Michigan State University Graduate Academic Conference, East Lansing, MI
- ***Zhang, B.**, Shevell, S. K., (June 2019). Binocularly-driven neural responses to chromatic interocular-switch rivalry. Oral Presentation, University of Chicago Graduate Conference, Chicago, IL

TEACHING EXPERIENCE

- Fall 2024 **Introductory Psychology**, Teaching Assistant
- Spring 2024 **Sensation & Perception | Cognitive Psychology**, Teaching Assistant
- Spring 2023 **Sensation & Perception | Cognitive Psychology**, Teaching Assistant
- Fall 2022 **Introductory Psychology**, Teaching Assistant
- Summer 2022 **Data Analysis in Psychological Research**, Instructor-of-Record
- Spring 2022 **Research Design and Measurements in Psychological Research**, Teaching Assistant
- Fall 2020 **Cognitive Neuroscience**, Teaching Assistant

PROFESSIONAL EXPERIENCE

- 2026-Present **Technical Assistant MSU Resource Center for Persons with Disabilities**
- 2021-2025 **Research Consultant Energize Colorado**
- 2018-2020 **Technical Assistant University of Chicago**
- 2017-2018 **Behavioral Technician Firefly Autism**
- 2015-2016 **Undergraduate Research Assistant Fostering Healthy Futures**

AWARDS & HONORS

Sep 2025	Dissertation Completion Fellowship, College of Social Sciences, Michigan State University	\$ 9,000
Jul 2025	Spotlight: Access Torch, College of Social Sciences, Michigan State University	
May 2025	Graduate Educator Advancement and Teaching Fellowship, The Graduate School, Michigan State University	\$ 4,500
Mar 2025	Educator of the Month, The Graduate School, Michigan State University	
Sep 2023	Conference Award, Council of Graduate Students, Michigan State University	\$ 300
Mar 2023	Research Enhancement Award, College of Social Sciences, Michigan State University	\$ 660
Jun 2018	MAPSS Scholarship, Division of Social Sciences, The University of Chicago	\$ 24,000
Nov 2017	Public Pathways Scholar, Dept. of Psychology, University of Denver	
May 2017	Academic Achievement and Citizenship Award, Dept. of Psychology, University of Denver	
May 2017	Academic Achievement Award, Dept. of Psychology, University of Denver	

OUTREACH & PROFESSIONAL DEVELOPMENT

SERVICE AND OUTREACH

2025	Fresh Check Day, Planning Committee Member
2025	Neurodiversity Fair, Planning Committee Member & Attendee
2025	University Health and Well-being Summit, Student Panelist
2024-Present	Graduate Student Accessibility and Support Network, Vice President
2024-Present	Disability Summit, Planning Committee Member & Student Speaker
2023-Present	Graduate Academic Conference, Judge
2022-2024	Psychology Scholars Undergraduate Program Workshop, Graduate Student Panelist
2022-2024	Council of Graduate Students, Student Representative
2022-2024	University Committee on Graduate Studies, Graduate Student Representative

DEVELOPMENT

May - Dec 2025	GREAT Fellowship: Graduate Resources for Accommodations and Supportive Practices, Michigan State University Graduate School	East Lansing, MI
Aug 2023	Systems Vision Science Summer School & Symposium, Max Planck Institute for Biological Cybernetics	Tübingen, Germany
Jun 2022	Certificate for College Teaching Workshop, Michigan State University	East Lansing, MI